Reproduction of Zhang, Li, Hu & Luo (IEEE T-ED 2009)

"Field Enhancement for Dielectric Layer of High-Voltage Devices on Silicon on Insulator"
IEEE Trans. Electron Devices **56**(10), pp. 2327–2334 — the ENDIF principle

Independent derivation, simulation and figure-by-figure comparison

Abstract

In an SOI high-voltage device the vertical breakdown is limited by the *silicon* critical field $E_{S,C}$, but the buried *dielectric* can sustain a much larger field E_I fixed by displacement continuity across the silicon/dielectric interface. The ENDIF (enhanced dielectric-layer field) principle raises E_I — and hence the vertical breakdown voltage $V_{B,V}$ — through three knobs: (1) a thin silicon layer (higher $E_{S,C}$), (2) a low-k buried dielectric, and (3) an interface sheet charge σ_{in}. This note derives the closed-form model and reproduces the paper's analytical figures (Figs. 2–5). All headline numbers match to within rounding.

Key results (model vs. paper)

Quantity	Model	Paper
$E_{S,C}$ at $t_S = 0.1\,\mu\text{m}$	$140.8\,\text{V}\,\mu\text{m}^{-1}$	$141\,\text{V}\,\mu\text{m}^{-1}$
E_I at $t_S = 1\,\mu\text{m}$, $\varepsilon_I = 3.9\varepsilon_0$, $\sigma_{\text{in}} = 0$	$161\,\text{V}\,\mu\text{m}^{-1}$	$161\,\text{V}\,\mu\text{m}^{-1}$
E_I at $t_S = 1\,\mu\text{m}$, $\varepsilon_I = 2.0\varepsilon_0$ (low-k)	$313\,\text{V}\,\mu\text{m}^{-1}$	$319\,\text{V}\,\mu\text{m}^{-1}$
E_I at $t_S = 0.1\,\mu\text{m}$	$422\,\text{V}\,\mu\text{m}^{-1}$	$423\,\text{V}\,\mu\text{m}^{-1}$
$V_{B,V}$ at $t_S = 1\,\mu\text{m}$	348 V	348 V
$V_{B,V}$ at $t_S = 0.1\,\mu\text{m}$	852 V	852 V
$V_{B,V}$ at $t_S = 1\,\mu\text{m}$, $\sigma_{\text{in}} = 4\times10^{12}\,\text{cm}^{-2}$	719 V	719 V

1 Theory

1.1 Displacement continuity: why the dielectric field is enhanced

At the silicon/dielectric interface the normal electric displacement is continuous up to any interface sheet charge σ_{in}:

$$\varepsilon_I E_I = \varepsilon_S E_{S,C} + q\sigma_{\text{in}} \quad\Longrightarrow\quad \boxed{E_I = \frac{\varepsilon_S}{\varepsilon_I}\,E_{S,C} + \frac{q\sigma_{\text{in}}}{\varepsilon_I}} \tag{1}$$

Because $\varepsilon_S/\varepsilon_I \approx 11.7/3.9 \approx 3$, the dielectric already carries $\sim 3\times$ the silicon field; lowering ε_I (low-k) or adding σ_{in} raises it further. This is the ENDIF idea: the blocking capability is set by how much field the *buried dielectric* can hold, not the silicon.

1.2 Silicon critical field vs. thickness

Breakdown in the (thin, fully depleted) silicon occurs when the ionization integral reaches unity. Carrying out that condition with the silicon Chynoweth coefficient and the thin-layer field profile yields the paper's compact fit (eq. 10), with t_S in micrometres:

$$E_{S,C}(t_S) = 9.783\,\frac{21.765 - \ln t_S}{\ln t_S + 3.975}\times 10^4 \;\;[\text{V cm}^{-1}], \tag{2}$$

which rises sharply for thin silicon — reaching $141\,\text{V}\,\mu\text{m}^{-1}$ at $t_S = 0.1\,\mu\text{m}$, far above the conventional constant $20\,\text{V}\,\mu\text{m}^{-1}$. The associated silicon critical ionization rate is the Chynoweth rate evaluated at $E_{S,C}$, $\alpha_{S,C} = A\,e^{-B/E_{S,C}}$ with $A = 7.03\times10^5\,\text{cm}^{-1}$, $B = 1.231\times10^6\,\text{V cm}^{-1}$ (Fig. 2).

1.3 Dielectric field and vertical breakdown voltage

Substituting $E_{S,C}(t_S)$ into the continuity relation gives $E_I(t_S, \varepsilon_I, \sigma_{\text{in}})$ (eq. 11). Integrating the field through the silicon ($\frac{1}{2}\varepsilon_I t_S$ effective) and the dielectric ($\varepsilon_S t_I$) gives the vertical breakdown voltage (eq. 13), with t_S, t_I in micrometres and $\varepsilon_S, \varepsilon_I$ relative:

$$V_{B,V} = \left\{ \left[9.783 \, \frac{21.765 - \ln t_S}{\ln t_S + 3.975} \right] \left(\varepsilon_S t_I + \tfrac{1}{2} \varepsilon_I t_S \right) + q \sigma_{\text{in}} t_I \times 10^{-4} \right\} \Big/ \varepsilon_I . \tag{3}$$

$V_{B,V}(t_S)$ is U-shaped: it rises for thin silicon (high $E_{S,C}$) and again for thick silicon (more voltage held in the silicon), with an optimum in between. Larger σ_{in} and smaller ε_I shift the whole family upward.

1.4 Enhancement factors

Relative to a conventional device ($E_{S,C} = 20\,\text{V}\,\mu\text{m}^{-1}$, $E_{I,\text{Con}} = 60\,\text{V}\,\mu\text{m}^{-1}$), the three knobs contribute additively, $\Delta\gamma = \Delta\gamma_1 + \Delta\gamma_2 + \Delta\gamma_3$ (eq. 12):

$$\Delta\gamma_1 = \frac{\varepsilon_S \, \Delta E_{S,C}}{\varepsilon_{SiO_2} E_{I,\text{Con}}}, \quad \Delta\gamma_2 = \frac{\Delta\varepsilon_I}{\varepsilon_{SiO_2} - \Delta\varepsilon_I}, \quad \Delta\gamma_3 = \frac{q\sigma_{\text{in}}}{\varepsilon_{SiO_2} E_{I,\text{Con}}}. \tag{4}$$

Each reaches the target $\Delta\gamma_i = 15.67$ (i.e. $E_I = 1000\,\text{V}\,\mu\text{m}^{-1}$) at $t_S \lesssim 50\,\text{nm}$, $\varepsilon_I = 0.25\varepsilon_0$, or $\sigma_{\text{in}} = 2 \times 10^{13}\,\text{cm}^{-2}$, respectively — so satisfying *any one* condition suffices. Interface charge is the most practical knob.

Parameters. $\varepsilon_S = 11.7$, $\varepsilon_{SiO_2} = 3.9$; $A = 7.03 \times 10^5\,\text{cm}^{-1}$, $B = 1.231 \times 10^6\,\text{V}\,\text{cm}^{-1}$; $t_I = 2\,\mu\text{m}$ (Figs. 3–5).

2 Figure-by-figure comparison

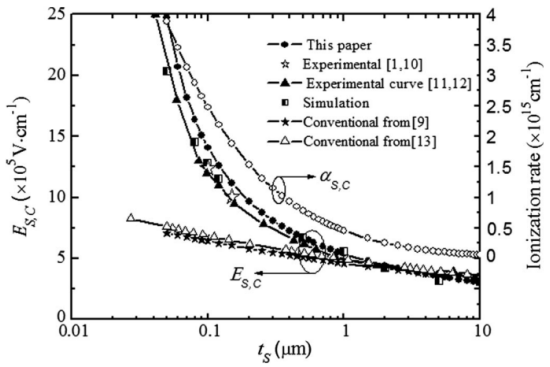
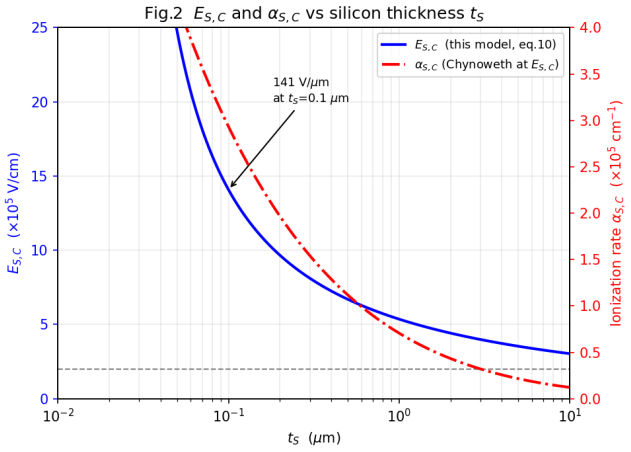


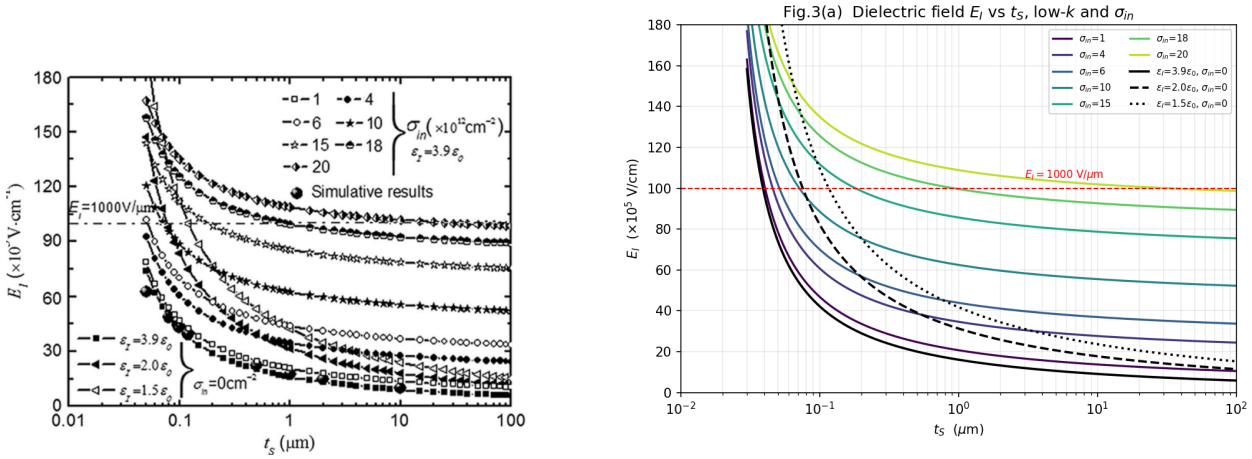
Fig. 2. Dependences of $E_{S,C}$ and $\alpha_{S,C}$ on t_S. Several analytical, simulative, and experimental results are compared. Results from [11]–[13] are obtained by

(a) Paper (original)



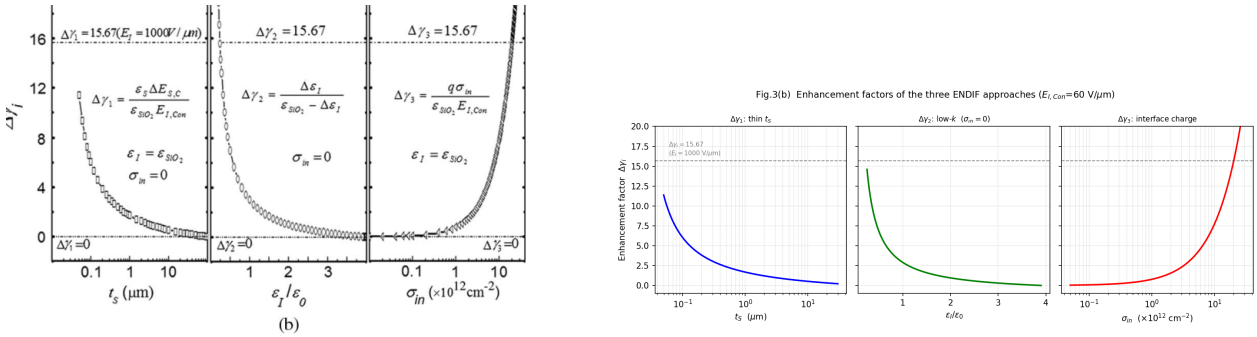
(b) This reproduction

Figure 1: **Fig. 2 — $E_{S,C}$ and $\alpha_{S,C}$ vs. t_S.** The critical field rises steeply for thin silicon ($141\,\text{V}\,\mu\text{m}^{-1}$ at $0.1\,\mu\text{m}$); the critical ionization rate $\alpha_{S,C}$ (right axis, Chynoweth at $E_{S,C}$) grows as t_S shrinks.



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Figure 2: **Fig. 3(a) — dielectric field E_I vs. t_S, low-k and σ_{in}.** Upper family: $\sigma_{\text{in}} = 1\text{–}20 \times 10^{12}\,\text{cm}^{-2}$ ($\varepsilon_I = 3.9\varepsilon_0$); lower black curves: $\varepsilon_I = 3.9/2.0/1.5\,\varepsilon_0$ ($\sigma_{\text{in}} = 0$). Both knobs push E_I toward the $1000\,\text{V}\,\mu\text{m}^{-1}$ dashed line.



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Figure 3: **Fig. 3(b) — enhancement factors $\Delta\gamma_i$.** Each approach reaches $\Delta\gamma_i = 15.67$ ($E_I = 1000\,\text{V}\,\mu\text{m}^{-1}$) on its own: thin t_S, low ε_I, or interface charge σ_{in} (crossing at $\sim 2 \times 10^{13}\,\text{cm}^{-2}$, matching the paper).

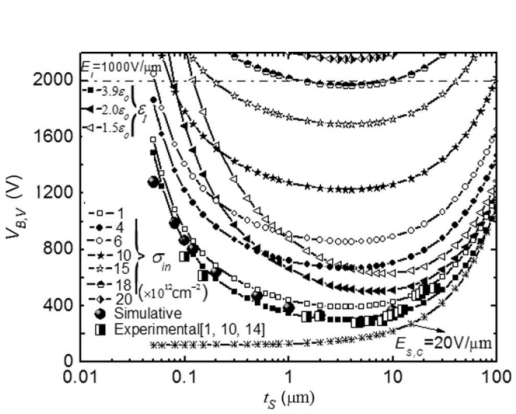
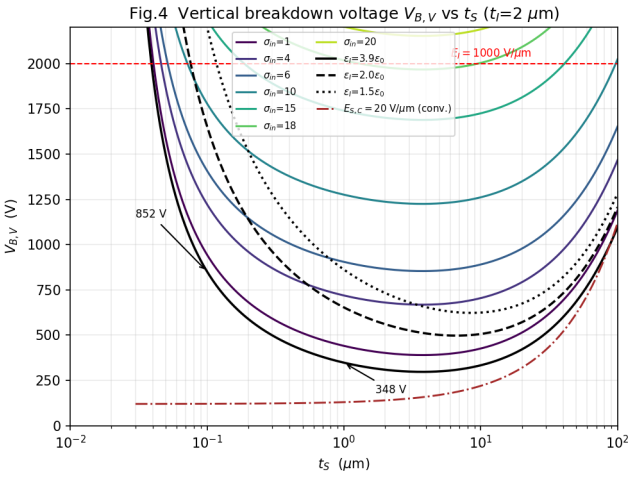
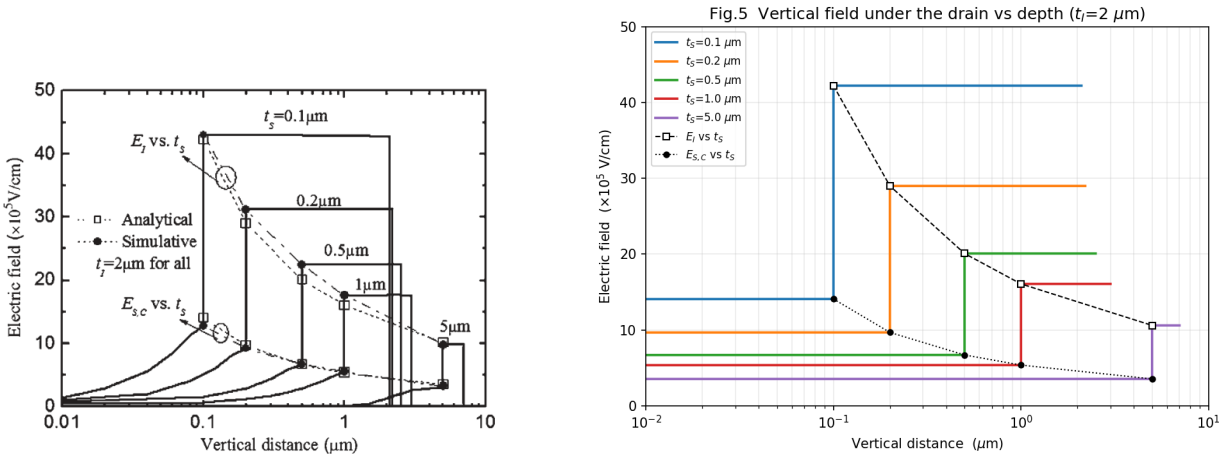


Fig. 4. Dependence of $V_{B,V}$ on t_S, low ε_I, and σ_{in} in the analytical, simulative, and experimental results. Linear impurity in the silicon layer is used while simulating, $N_{dn} = (1.5n + 1)N_{d0}$, where n is the number of



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Figure 4: **Fig. 4 — vertical breakdown voltage $V_{B,V}$ vs. t_S** ($t_I = 2\,\mu\text{m}$). U-shaped family; higher σ_{in} / lower ε_I lift $V_{B,V}$. The conventional $E_{S,C} = 20\,\text{V}\,\mu\text{m}^{-1}$ curve (bottom) is far lower for thin silicon. Model $V_{B,V} = 348/852\,\text{V}$ at $t_S = 1/0.1\,\mu\text{m}$ match the paper exactly.



(a) Paper (original)

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Figure 5: **Fig. 5 — vertical field under the drain vs. depth** for several t_S. Each profile shows the silicon field limited to $E_{S,C}$ then jumping to the plateau E_I across the buried dielectric; the two envelopes (E_I and $E_{S,C}$ vs. t_S) rise sharply for thin silicon.

3 Notes and limitations

What matches: the closed-form ENDIF quantities — $E_{S,C}(t_S)$, $\alpha_{S,C}$, E_I, $V_{B,V}$, and the enhancement factors — reproduce the paper's Figs. 2–5 and every stated number to within rounding.

Scope: this reproduction covers the paper's *analytical* core. Figs. 6–8 (step-thickness drift region, VK/LK SOI, and trench-SOI devices) are specific 2-D device structures compared against MEDICI simulations and experiment; reproducing those quantitatively requires a full 2-D device simulator and is out of scope here.

Erratum found in the paper's text: the sentence "when $t_S = 1\,\mu\text{m}$ and $\varepsilon_I = 3.9\varepsilon_0$, E_I values are 161 and $681\,\text{V}\,\mu\text{m}^{-1}$ for σ_{in} of 0 and $4 \times 10^{12}\,\text{cm}^{-2}$" is internally inconsistent: eq. (11) gives $E_I = 161 + 186 = 347\,\text{V}\,\mu\text{m}^{-1}$; the paper's own $V_{B,V} = 719\,\text{V}$ implies the same 347; and the paper's own Fig. 3(a) σ_{in}=4 curve reads $\sim 35 \times 10^5\,\text{V}\,\text{cm}^{-1}$ at $t_S = 1\,\mu\text{m}$. The origin of 681 is identifiable: computing with the *low-k* permittivity $\varepsilon_I = 2.0\varepsilon_0$ gives $319 + 362 = 681\,\text{V}\,\mu\text{m}^{-1}$ exactly — the LK value from the preceding sentence was carried into the interface-charge example. Text-level typo only; all equations and figures are mutually consistent (see `verify_681.py`).

Code: `zhang2009_full.py` — single file, closed-form model → Figs. 2–5, with a self-check that prints all the headline numbers above; `verify_681.py` — the three-way erratum verification.